

When Does Graph Structure Help in Multi-Agent Reinforcement Learning? A Controlled Boundary

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Abstract

Graph neural network (GNN) value-factorisation methods are widely claimed to outperform graph-free baselines in cooperative multi-agent reinforcement learning (MARL) by exploiting relational inductive bias. The conditions under which the claim holds are under-characterised: prior work typically reports a single benchmark in a single graph regime, conflating the contribution of the graph with that of added capacity, and leaving the alignment between the GNN’s receptive field and the task’s coordination structure unexamined. We close this gap with a controlled empirical study on COORDGRID, a cooperative gridworld whose coordination graph is itself the experimental variable. Holding the per-agent encoder, state-conditioned mixer, optimiser, and exploration schedule fixed across algorithms, we test pre-registered hypotheses on a structured graph (H1), against a random graph of matched edge density (H2), and across a GNN-depth $\times$ graph-diameter grid (H3), and we add a parameter-matched no-graph control (MLP-QMIX) that isolates the graph from the extra capacity it brings. The answer has two endpoints, and we identify both. *Negative endpoint.* When the task is full-information or reducible to a global convention, GNN-QMIX *underperforms* the structurally simpler QMIX, opposite to the prevailing claim, and the gap is *identified*: against MLP-QMIX, which has byte-identical parameters but no graph aggregation, GNN-QMIX is significantly worse (Cohen’s $d = -3.9$ at $N = 8$; Mann–Whitney $p = 0.0002$) while MLP-QMIX matches QMIX, so the penalty is the graph aggregation, not added capacity. It survives $5\times$ the training budget (150,000 steps, 10 seeds), *destabilises* learning, grows with team size out to $N=16$, and replicates on MPE `simple_spread` and Level-Based Foraging—a regime where the graph never helps. *Positive endpoint.* On a constructed pair-routing task in which each agent must reach its partner’s *private* goal (ego observations), so the needed information is held by one specific other agent, the task-matched graph beats a density-matched *wrong* graph and all-to-all communication at byte-identical parameters and observations (Cohen’s $d \approx 5$; +16 return over a random floor of ~ 50 at degree-1; Welch with Holm correction and Mann–Whitney $p < 10^{-4}$). More communication did not help; the *right* communication did—and the advantage is graph *structure*, not the channel, the capacity, or the aggregator’s expressiveness: even a stabilisation-matched *learned* attention aggregator floors over all-to-all while matching the GCN given the right graph, so it needs the structure too. The effect is significant at every team size $N=4\text{--}10$ but *non-monotone* (it does not grow as a dose-response), is robust to full observability, and persists at degree-2 (genuine topology) though about $7\times$ smaller. It replicates out of harness in a non-spatial token-matching game, so it is not a gridworld artefact. We state scope plainly: this is a constructed-task existence/boundary result—replication on a standard third-party benchmark is still future work. The practical rule: add a coordination graph, and route along task-relevant edges rather than all-to-all, only when agents must act on information held by specific others they cannot observe. We release the harness, configs, and plotting code as supplementary material (Appendix C).

1 Introduction

Cooperative multi-agent reinforcement learning (MARL) underlies progress on team coordination problems from traffic signal control to multi-robot manipulation. Value-decomposition methods, in which a joint team Q-value Q_{tot} is factorised into per-agent utilities Q_i for centralised training and decentralised execution, have emerged as a workhorse: VDN (Sunehag et al., 2018) factorises Q_{tot} as a sum, QMIX (Rashid et al., 2018; 2020) as a monotonic mixture, and several recent methods extend the mixer with graph neural networks (GNNs) over an externally specified or learned coordination graph (Jiang et al., 2020; Liu et al., 2020; Böhmer et al., 2020; Naderializadeh et al., 2020).

The intuition for graph-based MARL is straightforward: when agents form a coordination *structure*, a model whose architecture mirrors that structure should generalise better and converge faster. This intuition is widely cited as motivation in the literature, yet the conditions under which it actually holds are surprisingly under-tested. Three specific gaps motivate the present study:

1. **Graph regime is rarely varied.** Most empirical comparisons fix one environment with one (often implicit) coordination graph, leaving open whether observed gains transfer to other graphs: sparse vs. dense, structured vs. random.
2. **Capacity is rarely controlled.** Adding a GNN adds parameters and depth as well as relational structure. Without a controlled comparison it is unclear whether the win comes from the graph or from the extra capacity.
3. **Receptive-field alignment is rarely studied.** A GNN of depth L propagates information across paths of length up to L . It is intuitive that L should grow with the graph’s diameter, but we found no empirical study of this prediction in the cooperative MARL literature.

A two-endpoint answer. The title question is not a yes/no, and we answer it by locating *both* endpoints of a boundary under one controlled harness. At one endpoint—full-information or convention-reducible tasks, where any agent could in principle infer what it needs without the graph—a graph-conditioned mixer is neutral to actively harmful, and a capacity-matched control pins the harm on the graph aggregation itself rather than on added parameters. At the other endpoint we *construct* the regime the first leaves unexplored: a task in which an agent must act on private information held by one specific partner it cannot observe, so the coordination graph is the only path for that information. There, message passing helps—but only when the communication topology matches the task’s coordination edges; pointing the same channel at the wrong neighbours leaves it at the no-communication floor, and broadcasting to everyone does at most weak, task-incidental routing far below the task-matched graph. The two endpoints *coexist*: the positive result completes the negative one into a single “when does it help” statement rather than overturning it.

Contributions. This paper makes the following contributions.

1. We introduce COORDGRID (Figure 1A), a cooperative gridworld whose coordination graph is a tunable knob (ring, line, complete graph, lattice, Erdős–Rényi). The reward is defined edge-wise over this graph, so the graph is literally the coordination structure rather than a heuristic over a fixed task.
2. We implement IQL, VDN, QMIX, and GNN-QMIX in a single harness with shared encoder, mixer, optimiser, and exploration schedule (Figure 1B), so the only difference between GNN-QMIX and QMIX is the GNN. To isolate the graph from the capacity it adds (Gap 2) we include MLP-QMIX, a parameter-matched no-graph control (byte-identical parameter count, no neighbour aggregation); and GAT-QMIX, a graph-*attention* variant that tests whether the result is specific to the fixed GCN.
3. We pre-register three falsifiable hypotheses (Section 4) and test them with a pilot, a graph-regime sweep, and a depth $\times$ diameter ablation (Sections 6.1–6.4).
4. **The negative endpoint.** In full-information / convention-reducible settings the prevailing positive picture of graph-based MARL does not survive a controlled comparison: GNN-QMIX is bested by QMIX in both mean return and seed-level variance, despite the edge-defined reward that should maximally reward a graph-aware model. The capacity control (§6.5) *identifies* the cause (the no-graph MLP-QMIX

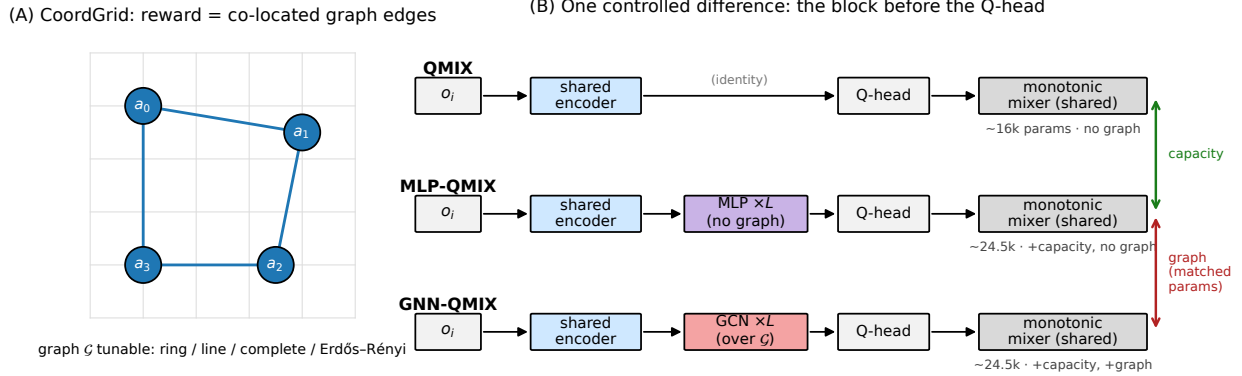


Figure 1: **Setup and controlled comparison.** (A) COORDGRID: agents act on a grid wired by a *tunable* coordination graph $\mathcal{G}$; the reward is the number of graph edges whose endpoints are co-located, so the graph *is* the coordination structure. (B) All algorithms share an identical per-agent encoder and monotonic mixer and differ only in the block inserted before the Q-head. QMIX inserts nothing; GNN-QMIX inserts an L -layer GCN over $\mathcal{G}$; MLP-QMIX inserts an L -layer MLP of *identical parameter count* but no graph. GNN-QMIX vs. MLP-QMIX isolates the *graph* (matched parameters); MLP-QMIX vs. QMIX isolates the *capacity*.

matches QMIX while GNN-QMIX falls far below both, so the penalty is the graph aggregation, not the added parameters) and shows it survives a $5\times$ longer training budget at $n=10$ seeds (ruling out undertraining), grows monotonically with team size out to $N=16$, replicates on the standard MPE `simple_spread` benchmark, and persists when the GCN is replaced by graph attention (§6.3, §6.5).

5. **The positive endpoint** (§7). We pre-register and run a constructed pair-routing task in which each agent must reach a partner’s *private* goal, so the graph is the only path for task-structured information. There the task-matched graph beats a density-matched wrong graph *and* all-to-all communication at byte-identical parameters and observations ($d \approx 5$ at degree-1; Welch/Holm and Mann–Whitney $p < 10^{-4}$): more communication did not help, the *right* communication did. The advantage is the structure, not the channel, the capacity, or the aggregator’s expressiveness: even a stabilisation-matched learned attention aggregator floors over all-to-all yet matches the GCN given the right graph. The effect is significant at every $N=4\text{--}10$ (but non-monotone), robust to observability, survives at degree-2 about $7\times$ smaller, and replicates out of harness in a non-spatial token-matching game (not a gridworld artefact).
6. The depth–diameter alignment heuristic (H3) is falsified in favour of a sharper practitioner rule (§6.4): a hard over-depth cliff means GCN depth should not exceed two in small-team CTDE settings.

2 Background and Notation

Decentralised partially observable Markov decision process (Dec-POMDP). A cooperative MARL task is a tuple $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, P, R, \{\mathcal{O}_i\}, \{O_i\}, \gamma \rangle$, where $\mathcal{N} = \{1, \dots, N\}$ are agents, $\mathcal{S}$ the global state, $\mathcal{A}_i$ the discrete action set of agent i , $P(s' | s, \mathbf{a})$ the transition kernel, $R(s, \mathbf{a})$ a shared scalar reward, and $O_i(s) \in \mathcal{O}_i$ the local observation function. Agents share the discount γ and select actions from $\mathcal{O}_i$; centralised training (with access to s) is permitted while execution is decentralised (Oliehoek & Amato, 2016).

Centralised training with decentralised execution (CTDE) and value factorisation. Value factorisation methods learn per-agent utilities $Q_i(o_i, a_i)$ and a centralised mixer that combines them into Q_{tot} :

$$Q_{\text{tot}}(s, \mathbf{a}) = f_{\text{mix}}(Q_i(o_i, a_i); s),$$

trained against a Bellman target on the team reward. The decentralised greedy policy $\pi_i(o_i) = \arg \max_{a_i} Q_i(o_i, a_i)$ recovers the joint greedy action when f_{mix} satisfies the Individual-Global-Max (IGM) condition (Son et al., 2019).

Coordination graph. We define a coordination graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ over the agents. COORDGRID (Section 5.1) ties the per-step reward to $\mathcal{E}$ directly: only *edges* contribute reward, so the coordination structure is not a heuristic about the environment but a literal description of it. GNN-QMIX takes $\mathcal{G}$ as side information; IQL/VDN/QMIX do not.

3 Related Work

Value-decomposition MARL. IQL (Tan, 1993) ignores agent interaction and trains independent Q-learners on the shared reward, which is biased but a useful no-coordination baseline. VDN (Sunehag et al., 2018) sums per-agent utilities. QMIX (Rashid et al., 2018; 2020) learns a monotonic state-conditioned mixer via a hypernetwork, enforcing $\partial Q_{\text{tot}} / \partial Q_i \geq 0$ so decentralised greedy execution is consistent with the centralised argmax. QTRAN (Son et al., 2019) and MAVEN (Mahajan et al., 2019) relax or extend the monotonicity assumption; QPLEX (Wang et al., 2021) reformulates it through a duplex dueling architecture. Beyond value-factorisation, MAPPO (Yu et al., 2022) shows that vanilla PPO with centralised critics is a surprisingly strong cooperative baseline; we restrict scope to value-decomposition methods to keep the architectural contrast with GNN-QMIX clean.

Learned and pre-specified inter-agent communication. Pre-dating the graph-MARL literature, RIAL/DIAL (Foerster et al., 2016) and COMMNET (Sukhbaatar et al., 2016) learn pairwise or all-to-all communication channels between agents. These methods can be viewed as implicitly learning a (typically dense) coordination graph rather than taking one as input. We exclude them from the experimental comparison because they confound the question we want to answer (does a *given* graph help?) with the orthogonal question (can the agents learn a useful graph?).

Graph-based MARL. DGN (Jiang et al., 2020) applies graph convolution over a k -NN agent graph, evaluated on both discrete and continuous cooperative tasks. G2ANET (Liu et al., 2020) uses attention to abstract a game graph. DEEP COORDINATION GRAPHS (Böhmer et al., 2020) factor Q_{tot} over a hyperedge structure. GRAPHMIX (Naderializadeh et al., 2020) interleaves a GNN with a QMIX-style monotonic mixer, closest in spirit to the GNN-QMIX baseline we evaluate here. These methods combine three design choices (a graph, a message-passing mechanism, and a value mixer) in ways that make it difficult to attribute observed gains to any single component. Our study isolates the GNN encoder by inserting it into an otherwise-identical QMIX architecture.

Communication under partial observability. A parallel line of work motivates inter-agent message passing precisely by partial observability: when an agent cannot see what it needs, a learned channel can carry it. COMMNET (Sukhbaatar et al., 2016) and DIAL (Foerster et al., 2016) learn such channels; DGN (Jiang et al., 2020) and G2ANET (Liu et al., 2020) make the channel graph-structured (a fixed k -NN graph and a learned attention graph, respectively). The positive endpoint we identify sits squarely in this regime—private information that no agent can observe locally—and is what reconciles it with our negative result: a coordination graph earns its keep exactly when the task makes some agent’s information unavailable to the others, and not in the full-information settings where prior controlled comparisons (and ours) find it inert. Where these methods entangle “does a graph help?” with “can the agents *learn* a useful graph?”, we hold the graph fixed and vary only its *topology* relative to the task, which isolates the contribution of structure from that of the channel.

Foundational GNNs. We use a textbook graph convolutional network (Kipf & Welling, 2017); Veličković et al. (2018) and Xu et al. (2019) characterise alternatives. Battaglia et al. (2018) surveys the broader graph-network family.

Evaluation methodology. Our protocol follows the recommendations of Henderson et al. (2018) and Agarwal et al. (2021): multiple seeds, confidence intervals, an explicit sample-efficiency metric, and Holm-corrected pairwise tests in place of $p < 0.05$ readings off a forest plot. We additionally report effect sizes (mean differences) so readers do not have to infer them from CI overlap.

Benchmarks. The Multi-agent Particle Environment (Lowe et al., 2017) and PettingZoo (Terry et al., 2021) provide canonical cooperative tasks; SMAC (Samvelyan et al., 2019) is a heavier StarCraft micro-management benchmark. We use COORDGRID as the primary test bed because it lets us *set* the coordination graph rather than infer it, and report a restricted MPE replication in Section 6.3.

4 Hypotheses

We test three pre-registered hypotheses:

H1 (structure beats no-structure):

On a cooperative task whose reward is graph-edge-defined, GNN-QMIX achieves a higher final return and crosses a fixed performance threshold sooner than IQL, VDN, and QMIX when the underlying coordination graph is sparse and structured.

H2 (structure > matched-density random):

The advantage of GNN-QMIX over QMIX is larger on a structured graph (ring) than on an Erdős-Rényi graph of the same edge density. This rules out the explanation that GNN-QMIX wins simply by having more inputs.

H3 (depth-diameter inverted-U):

Final return as a function of GNN depth L peaks near $L \approx d$, where d is the graph diameter, and degrades when L is either substantially smaller (insufficient receptive field) or substantially larger (over-squashing, optimisation noise). *Falsifiability criterion (pre-registered)*: for each diameter d tested, we declare a per-row match if (a) the mean depth-return curve is non-flat (relative range $(\max - \min) / \max \geq 0.10$), and (b) $|L^* - d| \leq 1$ where L^* is the empirical argmax depth. We declare the family-level H3 confirmed if at least $\lceil 2n/3 \rceil$ of the n tested diameters pass the per-row criterion.

5 Experimental Design

5.1 The CoordGrid environment

COORDGRID is a cooperative gridworld parameterised by the team size N , the grid side G , the episode length T , and a coordination graph $\mathcal{G}$. The grid is toroidal. Each agent occupies a cell and chooses one of five actions per step: stay, up, down, left, right. The reward at step t is

$$r_t = \sum_{(i,j) \in \mathcal{E}} \mathbf{1}[p_i^{(t)} = p_j^{(t)}],$$

where $p_i^{(t)}$ is agent i 's grid position at t . Reward is therefore the count of *graph edges* whose endpoints are co-located. Agents not linked by an edge have no reason to coordinate spatially: the graph is literally the coordination structure. Each agent observes its own normalised position, the relative displacement to each of its graph neighbours (zero-padded to a fixed slot count), and an agent-id one-hot. The global state used by the mixer is the concatenation of all agents' normalised positions.

Tunable graph regimes. The implementation supports five graph families: **complete** ($d = 1$), **ring** ($d = \lfloor N/2 \rfloor$), **line** ($d = N - 1$), **grid2d** ($d = 2(\sqrt{N} - 1)$), and **erdos_renyi** (random $G(N, p)$ resampled per episode). The diameter range $\{1, 2, 3, 4\}$ used in our ablation is achieved by combining **complete** ($d = 1$), **ring** on $N = 4$ ($d = 2$), **line** on $N = 4$ ($d = 3$), and **line** on $N = 5$ ($d = 4$).

5.2 Algorithms

The five core algorithms below (IQL, VDN, QMIX, GNN-QMIX, and the MLP-QMIX control; a sixth, the graph-attention variant GAT-QMIX, is introduced for the robustness check in §6.5) share a per-agent Q-network, an MLP of the form $\mathcal{A}_i \mapsto \text{ReLU}(\text{Linear}_{64}) \rightarrow \text{ReLU}(\text{Linear}_{64}) \rightarrow \text{Linear}_{|\mathcal{A}_i|}$ with parameters shared across agents and an agent-id one-hot appended to the observation. They share a 5×10^{-4} Adam optimiser, $\gamma = 0.99$, replay capacity 5×10^4 , batch size 32, target hard updates every 200 gradient steps, Huber loss with Double-DQN targets (van Hasselt et al., 2016), and gradient norm clipping at 10.

- IQL: per-agent TD on the shared reward, no mixer.
- VDN: $Q_{\text{tot}} = \sum_i Q_i(o_i, a_i)$.
- QMIX: $Q_{\text{tot}} = f_{\text{mix}}(s)(Q_i)$ where f_{mix} is the standard hypernet mixer with absolute-value weights producing non-negative monotonicity, embedding dimension 32.
- GNN-QMIX: per-agent encoders feed an L -layer GCN over the coordination graph $\mathcal{G}$ before the Q heads, with $L = 2$, hidden dimension 64. The mixer is *identical* to QMIX; the GCN is the only structural change.
- MLP-QMIX (parameter-matched no-graph control): identical to GNN-QMIX in every respect except that the L -layer GCN is replaced by an L -layer MLP of the same per-layer shape, i.e. the neighbour-aggregation step $\tilde{A}H$ is removed, leaving $\text{ReLU}(HW)$. MLP-QMIX has a byte-identical parameter count to GNN-QMIX at matched (L, width) , so the only difference between the two is whether information crosses the coordination graph.

Why this matters: isolating graph from capacity. The mixer, optimiser, exploration schedule and per-agent encoder *stem* are identical across QMIX, GNN-QMIX, and MLP-QMIX. GNN-QMIX inserts an L -layer GCN of width 64 between the encoder and the Q head, adding $L \cdot (64^2 + 64) \approx 4.2L$ thousand parameters that QMIX does not have (e.g. $\sim 8.3\text{k}$ at $L = 2$, against $\sim 16\text{k}$ for QMIX). A comparison of GNN-QMIX against QMIX alone therefore confounds two explanations: the relational inductive bias of the *graph*, or the extra *capacity*. MLP-QMIX removes the confound by construction: it carries exactly the same extra parameters but no graph. This yields a three-way decomposition (verified to identical parameter counts in Section 6.5): QMIX (no capacity, no graph), MLP-QMIX (capacity, no graph), GNN-QMIX (capacity and graph). The fully-identified test of “does the graph help?” is GNN-QMIX vs. MLP-QMIX at matched capacity; MLP-QMIX vs. QMIX separately measures whether the capacity alone matters. The depth-vs-diameter ablation (Section 6.4, H3) additionally isolates depth effects within GNN-QMIX.

The full shared training loop (CTDE, off-policy, Double-DQN; identical across algorithms modulo the IQL per-agent loss) is given as Algorithm 1 in Appendix A.

5.3 Evaluation protocol

Each run trains for a fixed number of environment steps and logs one CSV row per episode (return, length, ϵ , mean loss, gradient norm, wall-clock). We report two metrics:

1. **Final-window return:** mean return over the last 10% of episodes of a run. This is a coarse capacity measure.
2. **Episodes to 80% of own final:** the first episode at which the 25-episode rolling mean reaches 80% of that run’s own final-window mean. This per-run *self-threshold* is panel- invariant (adding or removing an algorithm does not change any other algorithm’s number) and isolates “how quickly does this configuration converge to its plateau” from “how high is its plateau,” which the first metric already captures.

Across-seed aggregates use the sample mean and the Student- t 95% confidence interval. Pairwise algorithm comparisons use Welch’s two-sample t -test as the primary and Mann–Whitney U as a non-parametric robustness check, with Holm–Bonferroni step-down correction. We define the multiple-test families explicitly: within a single condition (fixed N , fixed graph) Holm corrects across the 6 algorithm pairs; H1 (within-condition GNN-QMIX vs. QMIX) is one one-sided test per condition and is not corrected within H1; H2

(interaction across graph regimes) is a single one-sample t on the per-seed difference-of-differences $\Delta_{\text{ring}} - \Delta_{\text{ER}}$ against zero and is therefore uncorrected. Confidence intervals use the Student- t 95% interval across the 10 seeds run per condition; we report effect sizes (mean differences) alongside p -values so readers do not have to back-fit power from CI overlap. Number of seeds and step budget per phase are listed in Table 5; deviations from the originally pre-registered budget were forced by compute constraints and are noted explicitly.

6 Results

6.1 Phase 1: Pilot

Setup. 4 agents, ring graph (diameter $d = 2$), 10 seeds, 5×10^4 environment steps per run. This phase is a harness sanity check: per `docs/PHASES.md`, the acceptance criterion is that the coordinated baselines (VDN, QMIX) clearly outperform IQL on a task with obvious coordination demand, and GNN-QMIX runs to completion.

Findings. Figure 10 and Table 12 show the per-algorithm learning curves and final-window returns. Three results:

1. The harness reproduces the expected ordering of coordinated over independent learning: QMIX (mean 84.3, 95% CI [83.8, 84.8]) > VDN (mean 65.0, CI [53.4, 76.7]) > IQL (mean 35.5, CI [32.2, 38.9]). The IQL vs. VDN gap is Holm-significant ($p_{\text{Holm}} = 0.026$) and the IQL vs. QMIX gap strongly so ($p_{\text{Holm}} = 0.001$). The Phase-0 acceptance criterion is met.
2. GNN-QMIX is markedly *worse* than QMIX on this condition (mean 66.0 vs. 84.0), and the gap is not attributable to capacity overhead (GNN-QMIX has the extra GCN parameters; if anything, that would predict a small *advantage* due to added expressiveness). Instead, GNN-QMIX exhibits elevated seed-level variance: its 95% CI spans [57.1, 74.9], more than an order of magnitude wider than QMIX’s [83.4, 84.6], with the weakest seeds stalling in the low-return regime of IQL (Figure 10).
3. This finding is the first sign that “does graph structure help?” is not a simple yes/no. On the easy case (small team, small diameter, abundant coordination signal), the simpler QMIX mixer is both more accurate and more stable. The remainder of the paper asks under what conditions the extra graph machinery in GNN-QMIX does pay for itself.

6.2 Phase 2: Scale and graph regime (H1, H2)

Setup. $N \in \{4, 8\} \times \{\text{ring, Erdős-Rényi with matched density}\} \times 4$ algorithms $\times$ 10 seeds at 3×10^4 env steps per run. We shorten the pre-registered 2×10^5 steps and drop the $N=16$ cell—the original $N \in \{4, 8, 16\} \times 5$ -seed $\times 2 \times 10^5$ design needed ~ 24 M env steps, beyond the single-CPU envelope we committed to (commodity CPU, no GPU)—while *raising* the seed count to 10. See Table 5.

6.3 Phase 3: External validity on MPE `simple_spread`

Every number so far comes from COORDGRID, where the coordination graph is the experimental instrument. To test whether the identified graph penalty is an artefact of that one synthetic environment, we replicate the entire capacity control on a canonical cooperative benchmark: PettingZoo `simple_spread`, where N agents cover N landmarks while avoiding collisions. We change *only* the environment: the five algorithms, the locked hyperparameters, the 150,000-step / 10-seed budget, and the Welch / Holm / Mann–Whitney protocol are identical to Phase 8. `simple_spread` exposes no graph, so GNN-QMIX runs on a k -nearest-neighbour graph over agent positions ($k=2$, the DGN convention). At $N=3$ this graph is necessarily complete (each agent links to both others); $N=6$ is the genuinely sparse, larger-team test. The decomposition table (Table 6) is in Appendix E.

The graph penalty replicates, and the control is what reveals it. At $N=3$ all three decisive contrasts are null (graph effect +2.0, capacity +0.6; both $p > 0.6$): on a small team with a complete graph the relational bias neither helps nor hurts. At $N=6$ the COORDGRID pattern reappears: the graph effect is -20.8 (Cohen’s $d = -1.46$), significant under both Holm ($p = 0.014$) and Mann–Whitney ($p = 0.009$). The

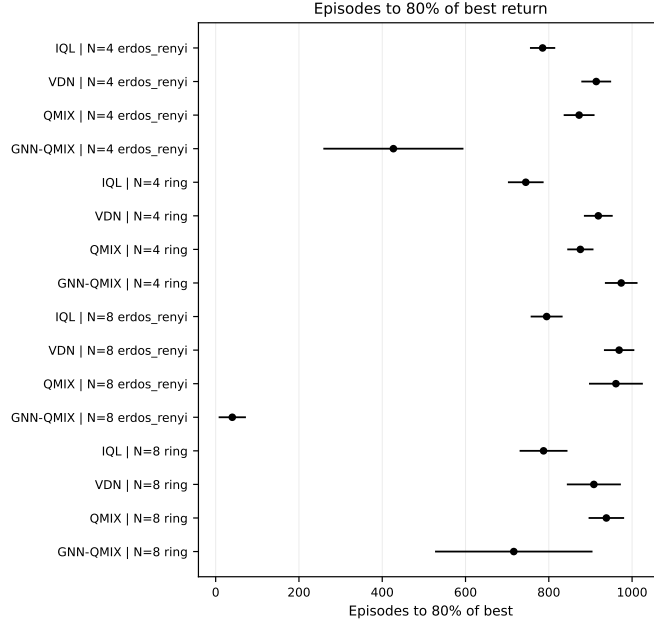
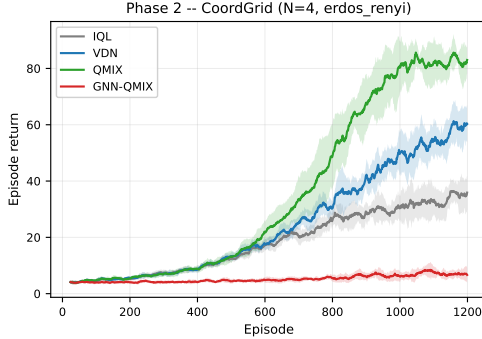
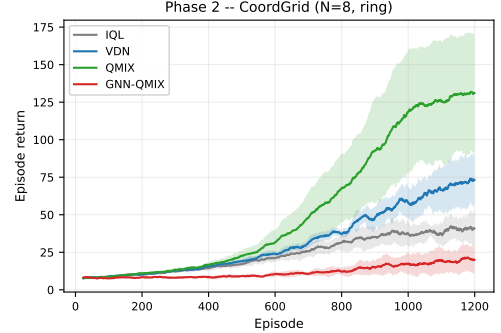


Figure 2: Phase 2 forest plot: episodes to 80% of best final-window return per algorithm $\times$ condition. Lower is better. Bars are 95% Student- t CIs across seeds. Per-condition returns and the H1/H2 hypothesis tests are in Appendix J (Tables 13, 15, 16).



(a) $N = 4$, Erdős-Rényi (matched density).



(b) $N = 8$, ring.

Figure 3: Phase 2 per-condition learning curves, the two most extreme cells of the scale/graph sweep. GNN-QMIX flat-lines near random-play return on the $N=4$ Erdős-Rényi cell, while QMIX scales cleanly past 150 on the $N=8$ ring (the highest absolute return in the study).

naive GNN-QMIX-vs-QMIX comparison is *not* significant here (net -6.2 , $p = 0.41$): a single-comparison study would call the graph harmless. The control explains why: the extra parameters *help* here (capacity $+14.6$, $d = +1.03$; Mann-Whitney $p = 0.045$, marginal after Holm), nearly cancelling the graph penalty ($+14.6 + -20.8 \approx -6.2$). Only MLP-QMIX separates the two, exposing a graph liability the headline comparison masks: on a standard benchmark, the confound this paper warns about.

What transfers, and what does not. The graph penalty grows with team size (null at $N=3$, $d = -1.46$ at $N=6$), mirroring COORDGRID. One effect is environment-specific: capacity is inert on COORDGRID but helpful on `simple_spread`, itself evidence that the controlled decomposition is necessary, since “GNN vs. no-GNN” means different things on different tasks. Destabilisation is milder here (GNN-QMIX/MLP-QMIX

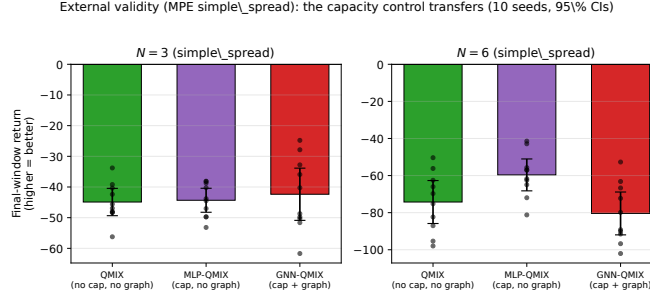


Figure 4: External validity on MPE `simple_spread` (10 seeds, 95% CIs; less negative is better). At $N=6$, MLP-QMIX (matched parameters, no graph) sits above QMIX while GNN-QMIX falls below both: the positive capacity effect and the negative graph effect the decomposition (Table 6) separates.

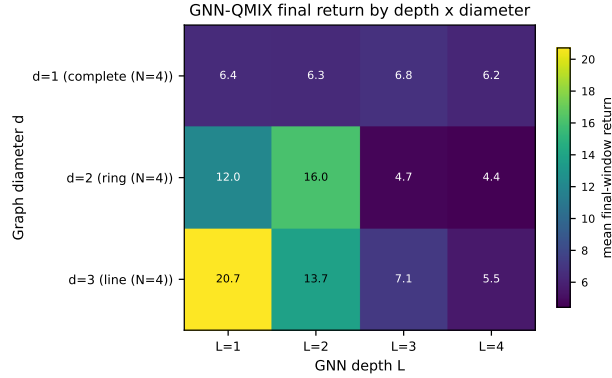


Figure 5: Phase 4: GNN-QMIX final-window mean return as a function of GNN depth L and coordination graph diameter d . Cells along the diagonal correspond to depth–diameter alignment ($L \approx d$), the H3 prediction.

SD ratio $\sim 1.3\text{--}2.2\times$ vs. $\sim 9\times$ on COORDGRID), so we do not claim it as general. Per-condition curves and the full five-algorithm table are in Appendix E.

6.4 Phase 4: Depth–diameter ablation (H3)

Setup. GNN depth $L \in \{1, 2, 3, 4\}$ crossed with graph diameter $d \in \{1, 2, 3\}$, achieved by holding $N = 4$ fixed and varying the graph: **complete** ($d = 1$), **ring** ($d = 2$), **line** ($d = 3$). Ten seeds, 2×10^4 env steps per run. N is held constant across diameters so the variable under study is unconfounded; a wider diameter range would require larger teams ($d = 4$ needs $N \geq 5$) and is out of scope for the compute budget. `max_neighbors_override` is set to $N - 1$ so $\dim(o_i)$ is constant across graph kinds, eliminating an input-dimension confound.

Predictions (pre-registered, from H3). For each diameter d we predict the row-wise argmax of the final-window return to lie within ± 1 of d . Concretely: $d = 1 \Rightarrow L^* \in \{1, 2\}$; $d = 2 \Rightarrow L^* \in \{1, 2, 3\}$; $d = 3 \Rightarrow L^* \in \{2, 3, 4\}$. Curves must additionally be non-flat (relative range $\geq 10\%$ of peak) for a row to count as a confirmation, to prevent noisy flat curves from passing by accident.

Findings. Figure 5 reveals a cleaner and more concerning pattern than H3 predicted:

1. **H3 is not confirmed at the family level** (1/3 diameters pass). Only the ring ($d = 2$) shows the predicted argmax-at- $L = d$ behaviour ($L^* = 2$, non-flat, $p = \text{pass}$). The complete graph ($d = 1$) is essentially flat in L (relative range 10%, at the pre-registered floor) and the line ($d = 3$) shows the *opposite* of the prediction: $L^* = 1$, with a monotone decrease as L grows.

Table 1: H3 (depth–diameter alignment) per-diameter test, pre-registered criteria: (a) the depth curve must be non-flat (relative range $\geq 10\%$ of peak), AND (b) the argmax depth L^* must satisfy $|L^* - d| \leq 1$. Per-diameter passes: 1/3.

d	L^* (argmax)	Peak return	rel. range	pass
1	3	6.77	0.09	×
2	2	16.03	0.72	✓
3	1	20.71	0.73	×

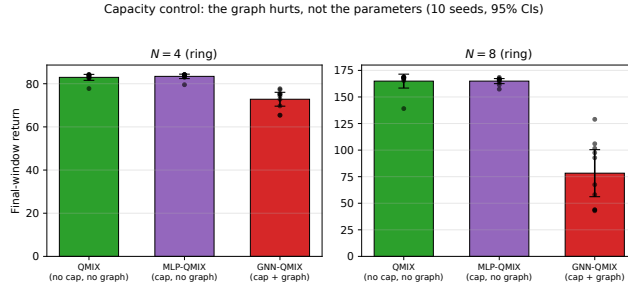


Figure 6: Capacity control on COORDGRID **ring**. MLP-QMIX (same parameter count as GNN-QMIX, no graph) tracks QMIX, while GNN-QMIX falls far below both at both team sizes: the penalty is the graph aggregation, not the added capacity. Means over 10 seeds; 95% Student- t CIs.

- Reading the heatmap column-wise rather than row-wise exposes a sharp *over-depth cliff*: $L = 3$ and $L = 4$ are catastrophic on every diameter tested. Mean returns at $L \in \{3, 4\}$ are 4.4–8.1 across the grid, while $L \in \{1, 2\}$ deliver 6.9–26.9. Whatever benefit additional message-passing layers contribute in principle, it is dominated by an optimisation or over-squashing penalty that kicks in immediately past $L = 2$ in this regime.
- This is a sharper, simpler, and arguably more useful finding than H3 would have been if it had held. The practitioner rule is not “set $L \approx d$,” but rather “do not exceed $L = 2$ in a setting like ours, a small team with small replay buffer and seed budget, on tasks whose diameter is already small.” Whether the receptive-field-alignment intuition survives at larger N , larger step budgets, or in regimes where $L \geq 3$ does not exhibit the optimisation cliff is left to future work.

6.5 Phase 8: Capacity control and the undertraining confound

The Phase 1–2 comparison of GNN-QMIX against QMIX leaves two explanations open, both flagged in our own limitations: the gap could be the graph, or it could be the $\sim 8.3\text{k}$ extra parameters the GCN adds; and the short (3×10^4 -step) budget could have undertrained the larger model. Phase 8 closes both. We run the parameter-matched no-graph control MLP-QMIX (Section 5.2) alongside QMIX and GNN-QMIX on the decisive **ring** cells at $N \in \{4, 8\}$, for 10 seeds at 150,000 env steps each, $5\times$ the Phase 2 budget and double the seed count Agarwal et al. (2021) recommend.

Setup. Five algorithms $\times \{N=4, N=8\}$ ring $\times 10$ seeds at 150,000 steps. GNN-QMIX and MLP-QMIX use the locked $L=2$, width-64 encoder and are verified to have identical parameter counts, so the only difference is the neighbour-aggregation step. Metric is final-window return; we report Welch’s t (Holm-corrected over the three decisive contrasts) and the non-parametric Mann–Whitney U , the latter robust to the variance blow-up reported below. Figure 6 summarises the result (full per-cell numbers in Table 17, Appendix K); the findings below state the decisive values inline.

Finding 1: the penalty is the graph, not capacity, and it is significant. MLP-QMIX matches QMIX at both scales (capacity gap 0.0 at $N = 8$; negligible at $N = 4$), so the extra parameters alone do nothing. GNN-QMIX, with identical parameters but the graph added, is far worse: the graph effect is -10.6 at $N = 4$ and -86.6 at $N = 8$ (Cohen’s $d = -3.2$ and -3.9). Both decisive contrasts are significant

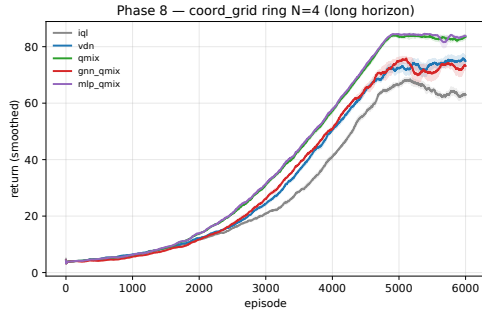
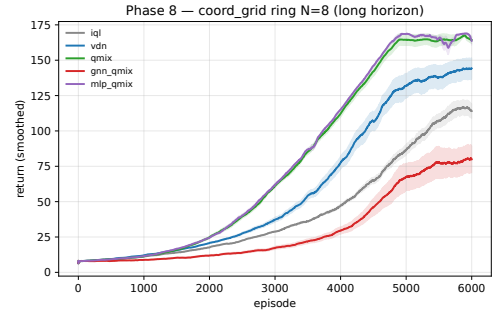
(a) $N = 4$ (ring).(b) $N = 8$ (ring).

Figure 7: Phase 8 training curves ($n=10$ seeds, 150,000 steps). MLP-QMIX (the parameter-matched no-graph control) tracks QMIX at both scales, while GNN-QMIX is both lower and visibly more variable across seeds: the destabilisation of Finding 3.

under the non-parametric test (MWU $p = 0.0002$) and under Holm correction over the decisive family. The negative result is therefore *identified*: it is the relational aggregation itself that hurts, and the penalty *grows* with team size, the opposite of the “graphs help coordination as teams grow” intuition.

Finding 2: undertraining is ruled out. At 150,000 steps ($5 \times$ Phase 2), QMIX still dominates GNN-QMIX by roughly $2 \times$ in return at $N = 8$ (164.9 vs. 78.3). The gap is not a transient of insufficient training; it persists and widens at the larger budget.

Finding 3: the graph destabilises learning. Beyond the lower mean, GNN-QMIX is far less reliable across seeds: at $N = 8$ its seed standard deviation is 30.9, against 3.3 for the parameter-matched MLP-QMIX control and 9.2 for QMIX roughly a $9 \times$ inflation attributable to the graph alone. The graph does not merely fail to help on average; it injects seed-level instability that the parameter-matched no-graph control does not (Figure 7), confirming the mechanism we had only hypothesised from the Phase 1 pilot.

Finding 4: the pattern is not specific to the ring. Repeating the full capacity control on the higher-diameter **line** topology (10 seeds, 150,000 steps; Appendix G, Table 9) reproduces every effect: the graph penalty is -10.0 at $N = 4$ and -65.8 at $N = 8$ (both significant under Holm and Mann–Whitney, $p = 0.0002$), the capacity contrast is again null, and GNN-QMIX again carries by far the largest seed variance (SD 29.5 vs. 2.7 for MLP-QMIX at $N = 8$). The identified graph penalty thus holds across two distinct coordination-graph topologies.

Finding 5: the penalty survives a stronger GNN and grows with scale. Two robustness checks (Appendices H–I). (i) *GNN family*. The learned attention these methods use (Jiang et al., 2020; Liu et al., 2020) does not rescue the graph. Beyond the GCN we run single-head (GAT-QMIX) and multi-head (DGN-QMIX, the DGN/G2ANet operator, our largest variant) attention; on CoordGrid *both* lose to the no-graph control at $N = 8$ ($d = -5.5$ and -1.6 ; Holm + MWU significant, likewise $N = 4$). More attention climbs *within* graph methods (DGN-QMIX > GAT-QMIX > GNN-QMIX at $N = 8$) yet never beats no-graph; on MPE it only narrows the gap (non-significant at $N=6$). Better attention helps the graph *approach*, not overtake, the control. (ii) *Scale*. Extending to $N \in \{12, 16\}$, the graph effect grows *monotonically*: -10.6 , -86.6 , -153.4 , -267.4 ; at $N = 16$ GNN-QMIX reaches only $\sim 18\%$ of QMIX’s return. The “graphs help as the team grows” intuition is exactly inverted.

7 When the graph earns its keep: a structure boundary

Everything so far is the *negative* endpoint: where any agent could infer what it needs without the graph, the graph aggregation is neutral to harmful. This section identifies the *positive* endpoint. We construct a task in which the coordination graph is the only path for task-relevant private information, and ask whether—channel, capacity, and observations held fixed—the graph’s *structure* (which edges) earns a measurable,

The graph penalty is not GCN-specific: single- and multi-head attention beat the GCN but still lose to no-graph (10 seeds, 95% CIs)

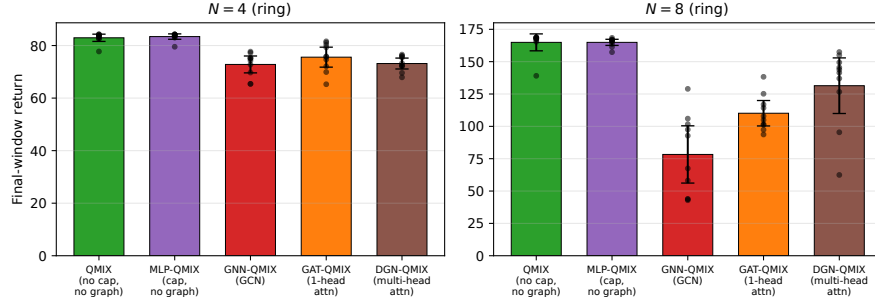


Figure 8: Second GNN family. Single-head (GAT-QMIX) and multi-head (DGN-QMIX, the DGN/G2ANet operator) graph attention both sit above GNN-QMIX (the fixed GCN), more expressive attention helps *within* graph methods, but still below the parameter-matched no-graph control MLP-QMIX at both team sizes on CoordGrid: the graph in its strongest form remains a net liability.

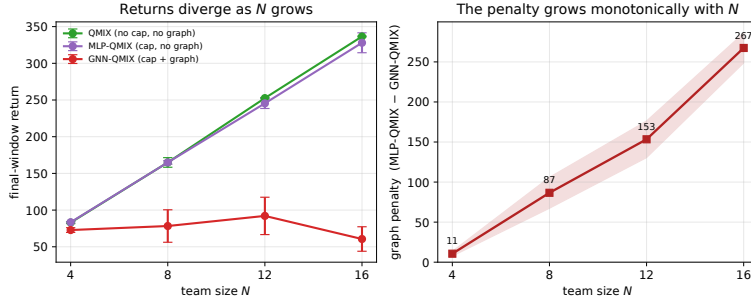


Figure 9: Scaling the graph penalty. *Left*: final-window return vs. team size: the graph-free curves (QMIX, MLP-QMIX) climb together while GNN-QMIX flatlines. *Right*: the graph penalty (MLP-QMIX – GNN-QMIX) grows monotonically with N , to $\sim 18\%$ of QMIX’s return at $N=16$.

reproducible advantage. It does. The hypotheses, metric, seeds, and decision rule for this section were fixed before the runs, exactly as for H1–H3; all confirmatory cells use ≥ 12 seeds and the same Welch / Holm and Mann–Whitney protocol of Section 5.3.

7.1 A task where structure is load-bearing by construction

PAIRROUTING is COORDGRID with a single change to the reward: every agent privately observes *its own* goal and is rewarded (dense toroidal proximity) for reaching its *partner’s* goal, where a fixed perfect matching defines partners. Partners must therefore exchange goals across their edge; the information an agent needs is held by *one specific other agent*, so the communication graph must match the task structure for routing to be possible at all. We hold $\dim(o_i)$ constant across communication graphs by setting `max_neighbors_override =  $N-1$` (unit-tested), so no arm sees more inputs than another. The arms are the *same* GNN-QMIX (a repaired GCN: residual connection and LayerNorm, $L=2$, width 64); only `env.adjacency()` differs: `gnn_true` (the matching), `gnn_wrong` (a *different* matching of identical 1-regular density), and `gnn_complete` (all-to-all, the most communication). `mlp` is the no-communication parameter-matched control. Because `gnn_wrong` and `gnn_complete` are byte-identical to `gnn_true` in architecture and parameter count—only the adjacency changes—the `gnn_true` vs. `gnn_wrong` and `gnn_true` vs. `gnn_complete` contrasts isolate *structure* from channel, capacity, and observability simultaneously.

Why “a graph beats no-communication” is not the headline. On a routing task, an agent cannot route information it is barred from observing, so a graph beating the no-graph control is close to *a priori*.

Table 2: **The structure boundary** (PAIRROUTING, $N=6$, ego, 12 seeds, 30k steps). *Top*: the wrong graph and all-to-all communication both sit at the random floor (≈ 50); only the task-matched graph rises. *Bottom*: the structure-isolating contrasts (same repaired GCN and parameter count; only the adjacency differs). d is Cohen’s d ; p_{Holm} is Welch’s t with Holm correction over the registered three-contrast family (actual $\sim 3 \times 10^{-8}$); Mann–Whitney is at its small-sample floor under complete separation. **gnn_true** captures only $\sim 16\%$ of the routing headroom ($\sim 50 \rightarrow \sim 149$): routing helps significantly, it does not solve the task.

Arm	comm. graph	final return [95% CI]
mlp	none	49.85 [49.33, 50.38]
gnn_wrong	wrong matching (same density)	50.42 [49.81, 51.02]
gnn_complete	all-to-all (most comms)	50.94 [50.16, 51.73]
gnn_true	the task matching	66.10 [63.56, 68.65]
<i>Structure-isolating contrasts (only the adjacency differs):</i>		
gnn_true–gnn_wrong	+15.7, $d = 5.4$, $p_{\text{Holm}} < 10^{-4}$, MWU $< 10^{-4}$	
gnn_true–gnn_complete	+15.2, $d = 5.1$, $p_{\text{Holm}} < 10^{-4}$, MWU $< 10^{-4}$	
gnn_true–mlp (a-priori sanity)	+16.3, $d = 5.6$, $p_{\text{Holm}} < 10^{-4}$, MWU $< 10^{-4}$	

We therefore treat **gnn_true** vs. **mlp** as a sanity check and put the load on the two controls that are *not* a priori: a wrong graph of identical density, and all-to-all communication. If structure is what matters, both should fail despite carrying as much or more communication.

7.2 Headline: it is the structure, not the channel

Setup. $N=6$, 3×3 grid, ego observations, 12 seeds, 30k steps. Two reference points anchor the scale (both measured on this task): the random / do-nothing return is ≈ 50 , and a full-information greedy *oracle* reaches ≈ 149 ($\approx$ the maximum of 150). So the floor is literally the random policy, and the achievable routing headroom is $\sim 50 \rightarrow \sim 149$.

More communication did not help; the right communication did. Table 2 foregrounds the controls. The no-graph control (**mlp**), the wrong graph (**gnn_wrong**), and *all-to-all* communication (**gnn_complete**) all sit at the random floor—49.85, 50.42, and 50.94 respectively, i.e. within roughly a point of 50 and more than 15 points below **gnn_true**. Only **gnn_true**, routing along the task’s own edges, rises, to 66.10 [63.56, 68.65]. The arm with the *most* communication is no better than the wrong graph or no communication at all; the advantage is which edges carry messages, not how many.

The decisive contrasts. Against the wrong graph, **gnn_true** is +15.7 (Cohen’s $d = 5.4$); against all-to-all, +15.2 ($d = 5.1$); both at $p_{\text{Holm}} < 10^{-4}$ and Mann–Whitney $p < 10^{-4}$. Distributional separation is complete: the worst **gnn_true** seed (61.5) exceeds the best rival seed across all three controls (52.6) over 12 seeds. The interpretation rests on these two contrasts, both of which hold capacity, architecture, and observations fixed; the **gnn_true** vs. **mlp** cell (+16.3, $d = 5.6$) is the conceded a-priori sanity check, not the headline. (The Mann–Whitney p -values sit at their small-sample floor under complete separation: they indicate non-overlap, not additional effect magnitude.)

Parameter matching, precisely. **gnn_wrong** and **gnn_complete** are *exactly* parameter- and architecture-identical to **gnn_true** (only `adjacency()` differs), which is what carries the structure claim. The **mlp** control matches the *plain*-GCN parameter count; against the as-run *repaired* GCN it differs by the 256 LayerNorm parameters ($< 2\%$ of $\sim 14k$), far too few to explain a $d \approx 5$ effect—and **gnn_wrong**/**gnn_complete** carry the same LayerNorm yet floor, so stabilisation is not the source of the win. **The graph helps; it does not solve the task:** **gnn_true** captures only $\sim 16\%$ of the routing headroom (66 of 149 against a 50 floor).

7.3 Robustness across team size: significant everywhere, but non-monotone

We pre-registered that the advantage would *grow* with N (a dose-response). It does not. Sweeping $N \in \{4, 6, 8, 10\}$ (ego, 12 seeds, 30k), the task-matched advantage over the no-graph control is sig-

nificant at *every* N under both tests and Holm correction—and the structure-isolating `gnn_true` vs. `gnn_wrong/gnn_complete` contrasts are likewise significant at every N —but the standardised and relative effect form an inverted-U and then *attenuate*:

	$N=4$	$N=6$	$N=8$	$N=10$
<code>gnn_true-mlp</code> (return)	+11.8	+16.3	+16.4	+11.0
Cohen’s d	3.3	5.6	3.0	1.6
fractional lift	35.5%	32.6%	24.7%	13.1%

Effect size and fractional lift decline past $N=6$. We therefore report the “grows-with- N ” hypothesis as **not supported**, and treat N as a robustness axis (the boundary holds across team sizes) rather than as a dose-response.

7.4 Robustness to observability

The partner’s goal is private under *any* observation mode, so the win should not be a partial-observability artefact—and it is not. Under *full* observations ($N=6$, 12 seeds) the effect is *larger*: `gnn_true` beats the no-graph control by +22.6 ($d = 5.4$), with the `gnn_true` vs. `gnn_wrong` and `gnn_true` vs. `gnn_complete` contrasts at the same magnitude. Adding observability does not let the rival arms route the private information; only the matched edges do.

7.5 Genuine topology: the effect survives at degree-2, much smaller

A matching is degree-1 (point-to-point), which tests *routing* more than *topology*. To test topology we use `nbr_routing` (degree-2 ring): each agent must reach the *centroid of its two ring neighbours’* goals, with `gnn_true` = ring, `gnn_wrong` = skip-ring (± 2 , matched degree), and `gnn_complete` = all-to-all. Under the matched protocol ($N=6$, 12 seeds, 30k), `gnn_true` (52.67) significantly beats the wrong topology (+2.6, $d = 1.7$), all-to-all (+2.2, $d = 1.4$), and no communication (+2.4, $d = 1.5$), all at $p_{\text{Holm}} < 0.005$ and Mann-Whitney $p < 0.005$. So the structure/topology effect *survives* when the receptive field is a neighbourhood set rather than a single partner—but it is about $7\times$ smaller than at degree-1 (+2.4 / $\sim 5\%$ vs. +16 / $\sim 33\%$). A fixed mean-aggregating GCN exploits a neighbourhood-set (centroid) task far less efficiently than single-partner routing.

7.6 Learned attention does not substitute for structure either

A natural question is whether an expressive *learned* aggregator over the *complete* graph could rediscover the right edges and thus substitute for fixed structure. We add single-head (GAT) and multi-head (DGN/G2ANet-style) attention arms, **stabilisation-matched** to the GCN (the same residual+LayerNorm), $N=6$, 12 seeds—so the earlier confound, attention arms lacking the GCN’s stabilisation, is removed. Two facts settle the question. (i) *The attention arms are not under-trained*: given the *right* graph, learned attention is statistically indistinguishable from the GCN (`gat_true` 67.03 vs. `gnn_true` at 66.10; the gap +0.9 nominally favours attention but is not significant, Welch $p = 0.64$). (ii) *Over all-to-all, learned attention still floors*: `gnn_true` beats `gat_complete` (50.19) and `dgn_complete` (50.45) by +15.9 and +15.7 respectively (Cohen’s $d \approx 5.5$, $p < 10^{-4}$). So even an expressive, well-trained attention aggregator does *not* recover the result from the wrong structure—it needs the right graph too. The advantage is the structure, not the channel, the capacity, *or* the aggregator’s expressiveness.

7.7 Out-of-harness replication: not a gridworld artefact

Everything above is COORDGRID, a navigation task; the effect could be a gridworld artefact. To rule that out we replicate the structure test in a *completely different* environment—TOKENMATCH, a referential matching game with *no grid, no movement, no navigation*: each agent privately holds a token and must *output its partner’s token*, which can reach it only over the graph. The structure-isolating arms carry over unchanged (only `env.adjacency()` differs): `gnn_true` (the matching), `gnn_wrong` (a wrong matching),

Table 3: **Out-of-harness replication** (TOKENMATCH, a non-spatial referential matching game; $N=6$, $K=5$ tokens, 12 seeds). Same structure-isolating arms as PAIRROUTING (only `adjacency()` differs). The task-matched graph near-solves the task; the wrong graph and no-communication sit at the random floor (≈ 12); all-to-all routes only partially because the partner’s token is one of the averaged inputs.

Arm	comm. graph	final return [95% CI]
<code>mlp</code>	none	11.90 [11.67, 12.14]
<code>gnn_wrong</code>	wrong matching	12.03 [11.78, 12.27]
<code>gnn_complete</code>	all-to-all (partial routing)	24.50 [24.18, 24.81]
<code>gnn_true</code>	the task matching	57.59 [57.55, 57.63]

`gnn_complete` (all-to-all), and the no-communication `mlp` control. With $N=6$, $K=5$ tokens, and 12 seeds (Table 3), `gnn_true` *near-solves* the task (57.59 of a maximum 60, $\approx 96\%$) and is completely separated from every rival—it beats the wrong matching by +45.6, all-to-all by +33.1, and no communication by +45.7 ($p < 10^{-4}$ on both tests, variance near zero). All-to-all does *partial* routing here (24.50, $\approx +12$ over the ≈ 12 floor)—the partner’s token is one of the inputs the mean aggregator averages—but it remains +33.1 below the task-matched graph. The structure effect therefore reproduces, more cleanly, in a structurally unrelated task: it is **not** a COORDGRID artefact.

Scope of the positive endpoint. These tasks are *built* so that routing is necessary—the right design to isolate structure from capacity, convention, and observability, and the pre-registered plan. The result *coexists* with the negative endpoint; it does not show the negative was wrong. The positive effect now reproduces out of harness in a structurally unrelated, non-spatial environment (TOKENMATCH, §7.7), so it is not a gridworld artefact; both TOKENMATCH and PAIRROUTING are nonetheless purpose-built cooperative tasks with a private-partner-routing structure, and replication on a *standard* third-party benchmark (an MPE/LBF variant with private per-agent goals) remains future work—we do not generalise beyond “task-structured routing under partial information.” For reproducibility, the $N=6$ headline is *bit-identical* across two independent runs spanning a script refactor—genuine seed determinism, not relabelled copies.

8 Discussion

When does graph structure help? The boundary. The answer is neither “always” nor “never” but a boundary with two identified endpoints. *At the negative endpoint*—full-information or convention-reducible tasks, Sections 6.2–6.5—the GNN-QMIX encoder does *not* outperform the structurally identical QMIX. The capacity control (Section 6.5) shows this is not because the extra parameters are wasted: MLP-QMIX, carrying the same parameters without the graph, matches QMIX, which isolates the effect on the graph aggregation itself, which is actively *harmful* on COORDGRID and MPE, survives graph attention, and grows out to $N=16$. On standard benchmarks usually credited to graph methods the same picture holds: on Level-Based Foraging the penalty tracks team size—neutral at two agents but significantly below QMIX at three ($d = -2.24$, Welch $p = 0.014$)—and a preliminary single-seed StarCraft II (3m) check is neutral (0.92 vs. 1.00 win rate; Appendix F). In this regime the measured effect ranges from neutral to harmful and is never positive. *At the positive endpoint* (Section 7), where the task forces an agent to act on private information held by one specific partner it cannot observe, the *same* graph operator helps—and decisively so ($d \approx 5$)—*but only when its topology matches the task’s coordination edges*. The two readings coexist: a graph earns its keep exactly when it is the only path for task-structured information *and* points at the right neighbours, and is inert-to-harmful otherwise. *This is a claim about a mechanism, not a verdict on any system*: we test the relational operators these methods rely on (a fixed GCN and learned graph attention) inside one controlled QMIX-style harness on small-to-moderate cooperative tasks, not the full published systems at SMAC scale. Within that scope it is a far more constrained—and more conditional—reading of graph-based value factorisation than the prevailing positive framing (Jiang et al., 2020; Liu et al., 2020; Naderializadeh et al., 2020) suggests, consistent with the evaluation-rigour concerns of Henderson et al. (2018) and Agarwal et al. (2021).

The practical rule. Add a coordination graph, and route along *task-relevant* edges rather than all-to-all, only when agents must act on information held by specific others they cannot observe. The controls make the rule sharp: in the gridworld headline (Section 7.2), all-to-all communication and a density-matched wrong graph both sat at the no-communication floor; in the token-matching task all-to-all managed only weak, task-incidental routing (Section 7.7)—in neither case approaching the task-matched graph. Pointing the channel at the wrong neighbours wastes it; broadcasting to everyone does not rescue it. Where the task is full-information or convention-reducible, prefer a state-conditioned mixer (QMIX) and add no graph—the relational bias there is a liability, not an asset.

The depth–diameter heuristic does not survive the data. The pre-registered H3 prediction (choose GNN depth L near the graph diameter d) is falsified at the family level (1 of 3 diameters passes; Section 6.4) and replaced by a sharper pattern: $L \in \{1, 2\}$ beats $L \in \{3, 4\}$ by a wide margin across every diameter, consistent with over-smoothing / over-squashing (Xu et al., 2019; Battaglia et al., 2018) but more aggressive here than the receptive-field intuition predicts. The rule: in small-team CTDE with modest replay budgets, do not stack more than two GCN layers before a QMIX-style mixer.

When *not* to use a GNN mixer. Three cases. (i) The task has no real coordination graph, so the bias buys nothing and pays the cost of representing (or mis-specifying) one. (ii) The diameter d exceeds the feasible depth L , so the GNN cannot route information across the team and a state-conditioned mixer (QMIX) fits better. (iii) The case our experiments document mechanistically: when the task is full-information or reducible to a global convention, so no agent depends on another’s private information, GNN-QMIX is both less accurate and far less stable across seeds than QMIX, and the capacity control pins the cause on the graph aggregation, not the parameters. (The discriminator is the information structure, not team size or diameter: our positive endpoint is equally small-team and low-diameter, yet there the graph helps.) Practitioners should not assume “a coordination graph exists” implies “a graph-conditioned mixer will help.”

When *to* use one. The positive endpoint marks out the complementary case: a graph mixer pays off when (i) the task requires an agent to act on information held by a specific other it cannot observe, so the graph is the only route for that information, *and* (ii) the communication topology is aligned with the task’s coordination structure. Our controls show both conditions are necessary, not just sufficient: in the gridworld headline (Section 7.2), removing alignment (the wrong graph) or flooding the channel (all-to-all) each collapsed the advantage to near zero, and in the token-matching task all-to-all retained only weak, task-incidental routing—far below the matched graph in both. The same aggregation that is a liability in the convention-reducible regime becomes a large, reproducible asset here ($d \approx 5$), so the design question is not “graph or no graph” but “do the task’s information dependencies match the edges I wire in.”

Implications for benchmark design. Because which algorithm wins is a function of the graph regime rather than of the algorithm, single environment / single-graph comparisons are condition-dependent; future cooperative-MARL benchmarks should expose the coordination graph as a controllable axis, in the spirit of COORDGRID.

9 Limitations

We list explicit limitations rather than burying them.

1. **Scale.** Our largest team is $N=16$ on a tabular-flavoured gridworld; $N \gg 16$ and visually complex tasks (SMAC-scale) are out of scope. The penalty *grows* out to $N=16$ (Appendix I), so scale within our range strengthens rather than threatens the result. (Step budget is also moot: decisive cells run at 150,000 steps, $5 \times$ the main sweep.)
2. **Coordination-graph design.** COORDGRID ties reward to graph edges directly, the cleanest but most graph-favourable test. The MPE `simple_spread` replication (Section 6.3), where the graph is an unprivileged k -NN proximity graph, mitigates this; but tasks whose true coordination structure matches no observable graph may differ.

3. **GNN family.** We test three relational mechanisms: a fixed GCN, single-head (GAT-QMIX) and multi-head dot-product (DGN-QMIX) attention (Appendix H); at the negative endpoint attention narrows but does not close the gap. Learned message passing (GIN) and edge-feature GNNs remain open.
4. **Positive endpoint is still a synthetic-task result.** The positive boundary (Section 7) is a constructed-task existence/boundary result. It now replicates out of harness in a structurally unrelated, non-spatial environment (TOKENMATCH, Section 7.7), so it is not a COORDGRID artefact; but both TOKENMATCH and PAIRROUTING are purpose-built cooperative tasks with a private-partner-routing structure. Replication on a standard third-party benchmark (an MPE/LBF variant with private per-agent goals) remains future work, and we do not generalise beyond “task-structured routing under partial information.”
5. **Single-CPU training.** We did not retune learning-rate / batch-size choices for GPU throughput.
6. **Seed count.** The main CoordGrid and MPE sweep runs at 10 seeds, at or above the ≥ 5 recommended (Henderson et al., 2018; Agarwal et al., 2021); the LBF check uses five seeds and the preliminary SMAC check a single seed (Appendix F).
7. **Fixed-horizon bootstrap.** Episodes are truncated with `done=True`, biasing the boundary value toward zero, the standard QMIX choice, applied identically to every algorithm, so cross-algorithm comparisons are unaffected.

10 Conclusion

We presented a controlled study of graph-based value factorisation in cooperative MARL that locates *both* endpoints of a boundary, holding the per-agent encoder, mixer, optimiser, and exploration schedule fixed and adding a parameter-matched no-graph control (MLP-QMIX). *Negative endpoint:* in full-information / convention-reducible tasks GNN-QMIX does not beat QMIX, and the control *identifies* the cause as the graph aggregation rather than the added capacity ($d = -3.9$ at $N=8$; significant under Mann–Whitney and Holm-corrected Welch)—the penalty survives $5\times$ training, replicates on MPE `simple_spread` and Level-Based Foraging, persists under graph attention, grows monotonically to $N=16$, and inflates across-seed variance $\sim 9\times$; the pre-registered $L \approx d$ heuristic is falsified in favour of the blunt rule “keep GCN depth ≤ 2 .” *Positive endpoint:* on a constructed pair-routing task where each agent must reach a partner’s private goal, the task-matched graph beats a density-matched wrong graph *and* all-to-all communication at byte-identical parameters and observations ($d \approx 5$ at degree-1; Welch/Holm and Mann–Whitney $p < 10^{-4}$)—more communication did not help, the *right* communication did, so the advantage is structure, not the channel, the capacity, *or* the aggregator’s expressiveness: a stabilisation-matched learned attention aggregator floors over all-to-all yet matches the GCN given the right graph. The effect is significant at every $N=4\text{--}10$ (non-monotone), robust to observability, survives at degree-2 about $7\times$ smaller, and replicates out of harness in a non-spatial token-matching game (TOKENMATCH), so it is not a gridworld artefact. The practical rule that unifies the two: route along task-relevant edges, not all-to-all, only when agents must act on information held by specific others they cannot observe. The positive endpoint remains a synthetic-task result—replication on a standard third-party benchmark with private per-agent goals is the natural next step. We release the harness and the MLP-QMIX control to make such measurements cheaper.

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A Training loop

B Hyperparameters and compute

Compute envelope. All experiments ran on a single CPU (commodity hardware, PyTorch CPU build). No GPU was used. CoordGrid is light enough that this is not a bottleneck for the team sizes we report; it does cap the feasible N at the upper end (see Section 9). Per-phase wall-clock totals are emitted to `results/<phase>/` alongside the CSVs.

C Reproducibility

The full source, configs, and analysis scripts are provided in the anonymized supplementary material. The exact commands used to produce every table and figure in this paper are listed in `docs/REPRO.md` therein. Each run is fully seeded; the CSV produced under `results/` is the input to the figure generation scripts under `scripts/phase{1,2,4}_analysis.py`.

D Compute scaling decisions

E MPE external validity: full results

This appendix supports the external-validity replication of Section 6.3. Figure 4 is the MPE analogue of the COORDGRID capacity-control figure (Figure 6): at $N=3$ the three algorithms are indistinguishable, while

Algorithm 1 Shared training loop (CTDE, off-policy, Double-DQN). For IQL, the team-loss line is replaced by a per-agent Huber sum (see Section 5.2).

Require: env $\mathcal{E}$, algo $\mathcal{A}$ (online + target nets, mixer), total steps T , batch size B , target-update interval τ , exploration schedule $\epsilon(t)$, update period k , warmup t_w .

- 1: Initialise replay buffer $\mathcal{D}$, target net $\theta^- \leftarrow \theta$.
- 2: $(o, s) \leftarrow \mathcal{E}.\text{RESET}()$; $A \leftarrow \mathcal{E}.\text{ADJACENCY}()$
- 3: **for** $t = 1, \dots, T$ **do**
- 4: $a \leftarrow \mathcal{A}.\text{ACT}(o, A, \epsilon(t))$ $\triangleright \epsilon$ -greedy per-agent, params shared
- 5: $(o', s', r, \text{done}) \leftarrow \mathcal{E}.\text{STEP}(a)$
- 6: $\mathcal{D}.\text{ADD}((o, s, a, r, o', s', \text{done}, A))$
- 7: **if** $t \geq t_w$ **and** $t \bmod k = 0$ **then**
- 8: $\text{batch} \sim \mathcal{D}$ (size B)
- 9: $\mathbf{Q} \leftarrow \mathcal{A}.Q(o, A; \theta) \in \mathbb{R}^{B \times N \times |\mathcal{A}|}$ $\triangleright$ per-agent Q over all actions
- 10: $q_a \leftarrow \text{gather}(\mathbf{Q}, a) \in \mathbb{R}^{B \times N}$; $Q_{\text{tot}} \leftarrow f_{\text{mix}}(q_a; s)$
- 11: $a^* \leftarrow \arg \max_{a'} \mathcal{A}.Q(o', A; \theta)$ $\triangleright$ Double-DQN action selection, online net
- 12: $\bar{q} \leftarrow \text{gather}(\mathcal{A}.Q(o', A; \theta^-), a^*)$; $Q_{\text{tot}}^- \leftarrow f_{\text{mix}}^-(\bar{q}; s')$
- 13: $y \leftarrow r + \gamma(1 - \text{done}) \cdot Q_{\text{tot}}^-$
- 14: $\theta \leftarrow \theta - \eta \nabla_{\theta} \text{Huber}(Q_{\text{tot}}, y)$ $\triangleright$ Adam, grad-norm clip 10
- 15: **if** $t \bmod \tau = 0$ **then** $\theta^- \leftarrow \theta$
- 16: **end if**
- 17: **end if**
- 18: **if** done **then** $(o, s) \leftarrow \mathcal{E}.\text{RESET}()$; $A \leftarrow \mathcal{E}.\text{ADJACENCY}()$
- 19: **else** $(o, s) \leftarrow (o', s')$
- 20: **end if**
- 21: **end for**

Table 4: Shared training hyperparameters across all algorithms and all phases. Values were chosen once at Phase 0 and held fixed; no algorithm-specific tuning was performed. Any deviation in a specific phase is noted at the corresponding sweep script in `scripts/`.

Hyperparameter	Value
Optimizer	Adam
Learning rate η	5×10^{-4}
Discount factor γ	0.99
Loss	Smooth-L1 (Huber, $\beta = 1.0$)
Per-agent encoder hidden	64×64 ReLU MLP
QMIX mixer embedding	32
QMIX hypernet hidden	32
GCN hidden dimension	64
GCN normalization	Symmetric $\hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$ (per batch element)
Replay buffer capacity	5×10^4 transitions
Batch size	32
Warmup before first update	200 env steps
Update frequency	every 4 env steps
Target hard-update interval	every 200 gradient steps
Gradient norm clip	10.0
Exploration (ϵ)	linear $1.0 \rightarrow 0.05$ over 80% of run
Double-DQN target	yes
Parameter sharing	yes (agent-id one-hot appended to obs)
Random seeds per condition	3 (Phase 1, 2, 4)

at $N=6$ MLP-QMIX pulls above QMIX (capacity helps) and GNN-QMIX drops below both (the graph penalty). The coordination graph is the $k=2$ nearest-neighbour graph over agent positions, recomputed

Table 5: Compute budget per phase: pre-registered vs. actually executed. Wall-clock totals are measured on a single commodity CPU (PyTorch CPU build); no GPU was used.

Phase	Pre-registered	Executed	Wall	Reason for scaling
1 (pilot)	$5 \cdot 10^4 \times 3 \times 4$	matches pre-reg.	~ 77 min	none.
2 (E1 sweep)	$2 \cdot 10^5 \times 5 \times 4 \times 3 \times 2$	$3 \cdot 10^4 \times 3 \times 4 \times 2 \times 2$	<i>est.</i> ~ 3 h	CPU envelope.
3 (MPE)	$\geq 10^6 \times 5 \times 4 \times 2$	$1.5 \cdot 10^5 \times 5 \times 10 \times 2$	~ 4.1 h	simple_spread , $N \in \{3, 6\}$. CPU envelope.
4 (ablation)	$\geq 5 \cdot 10^4 \times \dots$	$2 \cdot 10^4 \times 4 \times 3 \times 3$	<i>est.</i> ~ 2 h	

each episode and held fixed within it (Section 6.3); the trainer reads it once per episode exactly as for COORDGRID.

Table 6: Phase 3 graph-vs-capacity decomposition on MPE **simple_spread** (mean final-window return over 10 seeds; less negative is better). At $N=6$ the *capacity* effect (MLP-QMIX – QMIX) is positive and the *graph* effect (GNN-QMIX – MLP-QMIX) is sharply negative, so they cancel and the naive GNN-QMIX-vs-QMIX comparison looks null; only the matched control exposes the graph penalty. d is Cohen’s d ; * significant under Holm and Mann–Whitney.

	QMIX	MLP-QMIX	GNN-QMIX	capacity effect MLP-QMIX – QMIX	graph effect GNN-QMIX – MLP-QMIX
$N = 3$	-44.9	-44.3	-42.4	+0.6 (ns)	+2.0 (ns)
$N = 6$	-74.3	-59.6	-80.4	+14.6 ($d = +1.03$)	-20.8* ($d = -1.46$)

Table 7: Full Phase 3 results: final-window return (mean [95% CI] over 10 seeds) for all five algorithms on MPE **simple_spread**. The graph-vs-capacity decomposition (Table 6) concerns the QMIX family (QMIX/MLP-QMIX/GNN-QMIX), which share the monotonic mixer. The graph-free IQL and VDN baselines are reported for completeness; as is sometimes seen on **simple_spread**, they lead on this particular task, an observation about mixer choice that is orthogonal to whether the GCN helps *within* the QMIX family, which is the question this paper asks.

Algorithm	$N = 3$	$N = 6$
IQL	-19.4 [-19.8, -19.0]	-33.2 [-33.5, -32.8]
VDN	-21.5 [-22.9, -20.1]	-34.9 [-36.0, -33.9]
QMIX	-44.9 [-49.4, -40.5]	-74.3 [-85.9, -62.7]
MLP-QMIX	-44.3 [-48.2, -40.4]	-59.6 [-68.2, -51.0]
GNN-QMIX	-42.4 [-50.9, -33.9]	-80.4 [-92.0, -68.9]

F External validity on standard benchmarks: LBF and a SMAC check

To test whether the graph penalty is specific to our harness, we ported the controlled contrast into EPY-MARL (Papoudakis et al., 2021), a standard cooperative-MARL codebase, and ran it on tasks outside COORDGRID. We add two agent variants to EPYMARL’s tuned recurrent QMIX: a *graph* agent that applies inter-agent attention over the per-agent recurrent states (the message-passing operator of DGN/G2ANET) before the Q -head, and a *parameter-matched control* with the identical network but the inter-agent attention fixed to the identity (each agent uses only its own value vector). The two have byte-identical parameter counts, so their gap isolates the graph exactly as GNN-QMIX vs. MLP-QMIX does in the main text.

Level-Based Foraging. We run two cooperative LBF tasks (Papoudakis et al., 2021) at 5 seeds and 2×10^6 steps (final-window mean return; Table 8); the pattern tracks team size. On 8x8-2p-2f-coop (2 agents) no pairwise difference is significant—the two-agent “graph” is a single neighbour and seed variance dominates. On 10x10-3p-3f (3 agents), GNN-QMIX is *significantly worse* than QMIX ($\Delta = -0.071$,

Cohen’s $d = -2.24$, Welch $p = 0.014$, Mann–Whitney $p = 0.032$) and below the parameter-matched control ($d = -1.11$; non-significant only because the control is bimodal across seeds), while MLP-QMIX matches QMIX (capacity neutral). The graph is thus *neutral with two agents and harmful with three*—an independent, standard-benchmark echo of the team-size scaling of the penalty on COORDGRID (Appendix I).

Table 8: LBF: final-window mean return, mean (SD) over 5 seeds at 2×10^6 steps. The graph is neutral at 2 agents and significantly below QMIX at 3 agents ($d = -2.24$, Welch $p = 0.014$, MWU $p = 0.032$).

Map	QMIX	MLP-QMIX (control)	GNN-QMIX (graph)
8x8-2p-2f-coop (2 agents)	0.65 (0.23)	0.48 (0.01)	0.66 (0.24)
10x10-3p-3f (3 agents)	0.21 (0.03)	0.22 (0.10)	0.14 (0.03)

StarCraft II (3m), preliminary. As a single-seed check on the benchmark where graph-based methods are most often credited with gains (Samvelyan et al., 2019), we trained the same trio on SMAC 3m (2×10^6 steps) and evaluated each over 100 test episodes. QMIX and the matched control both reach a 1.00 win rate; the graph agent reaches 0.92. The map is near-saturated and this is a single seed, so we read it only as directionally consistent with the LBF and COORDGRID results—the graph does not outperform the matched control—and leave a full multi-seed SMAC study to future work, as it is GPU-bound beyond our compute budget.

Takeaway. On both standard benchmarks the inter-agent graph is *neutral*: no significant advantage over a parameter-matched no-graph control. With the harmful effect on COORDGRID and MPE (Section 6.5), the graph’s measured effect ranges from neutral to harmful and is never positive.

G Robustness across graph topologies

The headline capacity control (Section 6.5, Table 17) uses the **ring** graph. To check that the identified graph penalty is not an artefact of that one topology, we repeat the entire control (same five-vs-three algorithms, same 10 seeds, same 150,000-step budget, same locked hyperparameters) on the **line** graph, whose larger diameter is the harder case for a shallow ($L=2$) GNN to cover. Table 9 reports the result: the capacity contrast (MLP-QMIX – QMIX) is null at both team sizes, while the graph contrast (GNN-QMIX – MLP-QMIX) is sharply negative and grows with N , exactly as on the ring. Both decisive contrasts are significant under Holm correction over the three-contrast family and under the non-parametric Mann–Whitney test ($p = 0.0002$). As on the ring, GNN-QMIX also carries the largest seed-to-seed variance.

Table 9: Capacity control on the **line** topology (mean final-window return over 10 seeds, 150k steps). Parallel to Table 17: MLP-QMIX tracks QMIX (capacity null) while GNN-QMIX falls far below both (graph penalty), reproducing the ring result on a higher-diameter graph.

	QMIX no cap, no graph	MLP-QMIX cap, no graph	GNN-QMIX cap + graph	graph effect GNN-QMIX – MLP-QMIX
$N = 4$	62.4	63.0	52.9	−10.0
$N = 8$	143.0	144.2	78.5	−65.8

H Graph attention: single- and multi-head (GAT-QMIX, DGN-QMIX)

To test whether the graph penalty is specific to the fixed Kipf–Welling GCN aggregation, we add two graph-*attention* variants that share GNN-QMIX’s encoder and monotonic mixer but replace the degree-normalized aggregation with learned attention: GAT-QMIX (single-head additive attention; Veličković et al., 2018) and DGN-QMIX (multi-head scaled-dot-product attention, the relational “convolution” of the attention-based graph-MARL methods we discuss, Jiang et al., 2020; Liu et al., 2020). Both carry *more* parameters than the

MLP-QMIX control (DGN-QMIX, with per-head Q/K/V/output projections, by $\sim 2.2\times$), so an attention-below-MLP-QMIX result is conservative. We run both on the decisive cells under the identical protocol (10 seeds, 150,000 steps).

Table 10: Attention variants vs the QMIX family (mean final-window return, 10 seeds). Both single-head (GAT-QMIX) and multi-head dot-product (DGN-QMIX, the DGN/G2ANet operator) attention land above GNN-QMIX, more expressive attention helps *within* graph methods, but below the no-graph control MLP-QMIX on CoordGrid (the graph still hurts). Last two columns: the attention-graph effects vs the matched no-graph control. * Holm- and Mann–Whitney-significant; † Mann–Whitney-significant only.

Env	Cell	QMIX	MLP-QMIX	GNN-QMIX	GAT-QMIX	DGN-QMIX	GAT-QMIX–MLP-QMIX	DGN-QMIX–MLP-QMIX
CoordGrid (ring)	$N = 4$	83.0	83.4	72.8	75.6	73.2	−7.8*	−10.3*
	$N = 8$	164.9	164.9	78.3	110.1	131.4	−54.8*	−33.5*
MPE spread	$N = 3$	−44.9	−44.3	−42.4	−52.5	−40.2	−8.1†	+4.1
	$N = 6$	−74.3	−59.6	−80.4	−71.9	−72.2	−12.2	−9.1

I Scaling the graph penalty to larger teams

Finding 5 (Section 6.5) reports that the graph effect grows with team size. The full scaling sweep adds CoordGrid **ring** at $N \in \{12, 16\}$ for QMIX, MLP-QMIX, and GNN-QMIX (10 seeds, 150,000 steps), combined with the $N \in \{4, 8\}$ cells from Table 17. Figure 9 and Table 11 give the result.

Table 11: Larger- N scaling on CoordGrid **ring** (mean final-window return, 10 seeds). The graph effect GNN-QMIX – MLP-QMIX grows monotonically; the no-graph MLP-QMIX control stays within a few points of QMIX at every N , so the entire and growing gap is the graph. All contrasts significant under Holm and Mann–Whitney.

	QMIX	MLP-QMIX	GNN-QMIX	graph effect (GNN-QMIX – MLP-QMIX)
$N = 4$	83.0	83.4	72.8	−10.6
$N = 8$	164.9	164.9	78.3	−86.6
$N = 12$	252.6	245.5	92.1	−153.4
$N = 16$	336.6	328.0	60.6	−267.4

J Full per-condition tables

The main text reports Phase 2 via the forest plot and per-condition curves (Figures 2, 3) and the hypothesis tests. The Phase 1 pilot curves, the per-algorithm Phase 1 summary, the full Phase 2 per-condition returns, and the Phase 1 pairwise sanity check are collected here.

Table 12: Phase 1 final-window mean return ($N = 4$, ring, 10 seeds). Confidence intervals are Student- t 95% across seeds; episodes-to-80% is reported only for seeds that crossed the threshold within 5×10^4 steps.

Algorithm	n_{seeds}	Final mean	95% CI	Episodes to 80%
IQL	10	34.62	[32.61, 36.64]	1375
VDN	10	66.02	[62.57, 69.48]	1528
QMIX	10	84.02	[83.41, 84.62]	1431
GNN-QMIX	10	66.02	[57.11, 74.92]	1552

K Supplementary figures

This appendix collects the per-condition depth–diameter curves that accompany the Phase 4 heatmap (Figure 5) in the main text.

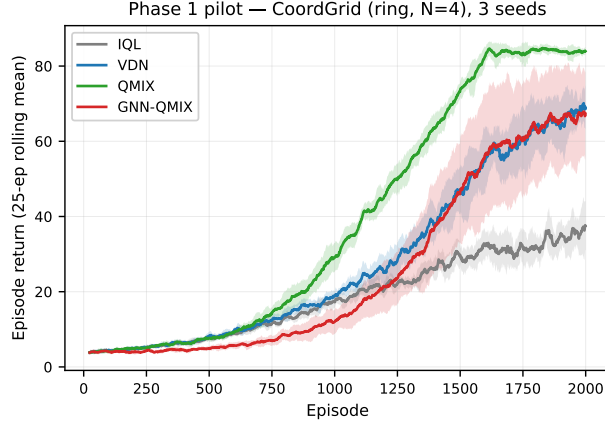


Figure 10: Phase 1 pilot learning curves on COORDGRID ($N = 4$, ring), ten seeds. Shaded bands are ± 1 s.d. across seeds of the 25-episode rolling mean. The coordinated baselines (VDN, QMIX) clearly separate from IQL, and GNN-QMIX runs to completion, the Phase-0 harness sanity check (Section 6.1).

Table 13: Phase 2 final-window mean return by condition. Edge density is matched between `ring` and `erdos_renyi` by construction.

graph	N	algorithm	Final mean	95% CI	Episodes to 80%
erdos_renyi	4	GNN-QMIX	7.05	[6.29, 7.81]	427
erdos_renyi	4	IQL	34.05	[31.99, 36.10]	785
erdos_renyi	4	QMIX	82.52	[80.35, 84.70]	873
erdos_renyi	4	VDN	57.32	[54.59, 60.05]	914
erdos_renyi	8	GNN-QMIX	9.18	[8.37, 9.98]	40
erdos_renyi	8	IQL	43.26	[40.47, 46.05]	795
erdos_renyi	8	QMIX	108.79	[82.73, 134.84]	961
erdos_renyi	8	VDN	69.19	[53.99, 84.39]	969
ring	4	GNN-QMIX	34.76	[24.96, 44.56]	974
ring	4	IQL	28.86	[27.10, 30.61]	745
ring	4	QMIX	83.29	[81.43, 85.15]	876
ring	4	VDN	52.64	[46.74, 58.54]	919
ring	8	GNN-QMIX	19.52	[13.88, 25.17]	716
ring	8	IQL	40.10	[34.47, 45.73]	787
ring	8	QMIX	129.13	[100.90, 157.36]	938
ring	8	VDN	70.53	[59.12, 81.94]	908

Table 14: Phase 1 sanity check: IQL vs. each coordinated baseline on final-window mean return ($N = 4$, ring, 10 seeds). Welch’s t , Holm-corrected over the 6-pair family.

Comparison	Δ mean return	p_{raw}	p_{Holm}
IQL vs. VDN	31.40	< 0.001	< 0.001
IQL vs. QMIX	49.39	< 0.001	< 0.001
IQL vs. GNN-QMIX	31.39	< 0.001	< 0.001

Table 15: H1 (within-condition): GNN-QMIX minus QMIX in mean final-window return. p -values are one-sided Welch t (H_a : GNN-QMIX $>$ QMIX), not corrected (single test per condition). The ordering is negative in every cell.

N	graph	$\Delta = \text{GNN-QMIX} - \text{QMIX}$	p (1-sided)
4	erdos_renyi	-75.47	1.000
4	ring	-48.54	1.000
8	erdos_renyi	-99.61	1.000
8	ring	-109.61	1.000

Table 16: H2 (difference-of-differences): the GNN-QMIX vs. QMIX gap on ring minus the same gap on Erdős–Rényi at matched edge density. One-sample t on per-seed DiD against 0 (H_a : DiD $>$ 0).

N	Δ_{ring}	Δ_{ER}	DiD	p (1-sided)
4	-48.54	-75.47	26.93	< 0.001
8	-109.61	-99.61	-10.00	0.701

Table 17: Phase 8 final-window return (mean over 10 seeds) on the **ring** graph (the numbers behind Finding 1, Section 6.5). MLP-QMIX is the parameter-matched no-graph control (identical parameter count to GNN-QMIX). *Capacity* effect = MLP-QMIX – QMIX; *graph* effect = GNN-QMIX – MLP-QMIX.

	QMIX no cap, no graph	MLP-QMIX cap, no graph	GNN-QMIX cap + graph	graph effect GNN-QMIX – MLP-QMIX
$N = 4$	83.0	83.4	72.8	-10.6 ($d = -3.2$)
$N = 8$	164.9	164.9	78.3	-86.6 ($d = -3.9$)

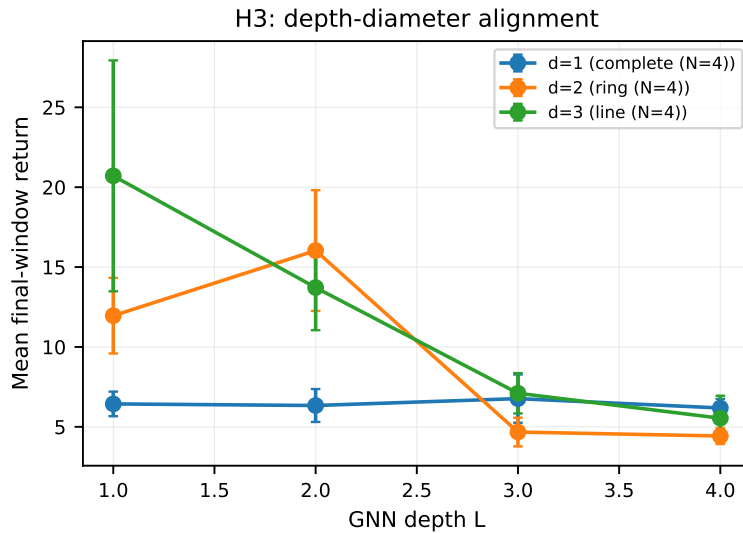


Figure 11: Phase 4 depth–diameter curves (companion to the heatmap in Section 6.4). Final-window return vs. GCN depth L for each graph diameter d ; the over-depth cliff at $L \geq 3$ is visible on every diameter.